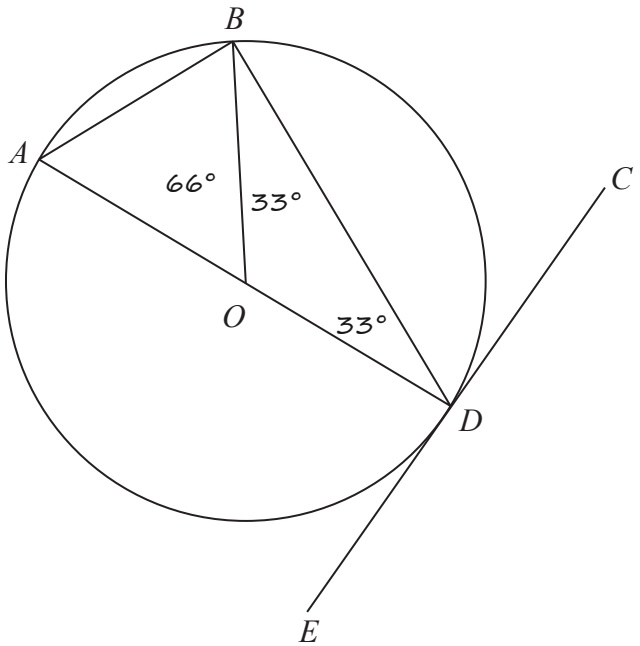
17



A, *B* and *D* are points on the circumference of a circle centre *O*.

EDC is a tangent to the circle.

Angle *BDC* = 57°

Find the size of angle *AOB*.
You must give a reason for each stage of your working.

OD is perpendicular to tangent DC.

Angle ODB = 90° – 57° = 33°.

1 mark
Find angle ODB

OB = OD, so angle OBD = 33°.

Angle BOD = 180° – 33° – 33° = 114°.

1 mark
Complete method

A, O, D lie on a straight line.

Angle AOB = 180° – 114° = 66°.

2 marks total
66° with reasons
Radius perpendicular to tangent; isosceles; triangle; straight line

(Total for Question 17 is 4 marks)